

# GEORGETOWN UNIVERSITY

## DEPARTMENT OF CHEMISTRY

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Matthew Cotaj

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SYNTHESIS AND PHASE CONTROL OF LANTHANIDE CHALCOGENIDE BINARY AND

MIXED-VALENT DOPED LANTHANIDE COPPER CHALCOGENIDE TERNARY

MATERIALS FOR TUNABLE MAGNETIC CHARACTERISTICS

submitted to the faculty of the Department of Chemistry

in partial fulfillment of the requirements for the degree of

**Bachelor of Science in *Chemistry* with Honors**

This thesis is complete and satisfactory in all respects, and any and all revisions required

by the final mentor and reader have been made.

Thesis Advisor



Signature

6 May 2026

Date

Reader



Signature

6 May 2026

Date

Undergraduate Honors  
Coordinator

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Signature

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Date

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AND  
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Matthew A. Cotaj

Thesis Advisor: Sarah Stoll, Ph.D.

**ABSTRACT**

The synthesis and phase control of lanthanide chalcogenide nanocrystals present an enduring challenge for colloidal synthesis owing to the fundamental mismatch between the hard-acid character of trivalent lanthanide ions and the soft-base character of chalcogenide anions. This thesis reports the colloidal synthesis of phase-pure lanthanide sulfide binary nanoparticles ( $\text{Nd}_3\text{S}_4$ ,  $\text{Gd}_3\text{S}_4$ ) via systematic screening of lanthanide halide precursors and sulfur sources, as well as mixed-valent lanthanide copper chalcogenide ternary nanorods ( $\text{Eu}_2\text{CuS}_3$ ,  $\text{EuGdCuS}_3$ ) using single-source dithiocarbamate (DTC) precursors in oleylamine under inert conditions. Phase purity was confirmed by powder X-ray diffraction (PXRD); composition was established by inductively coupled plasma mass spectrometry (ICP-MS) and energy-dispersive X-ray spectroscopy (EDS); and morphology was characterized by transmission electron microscopy (TEM) and high-resolution TEM (HRTEM). Magnetic properties were measured by superconducting quantum interference device (SQUID) magnetometry.  $\text{Eu}_2\text{CuS}_3$  nanorods exhibit a ferromagnetic transition at  $T_C = 3.6$  K and a saturation magnetization of  $M_{\text{sat}} = 47.8$  emu/g, consistent with the ordered  $\text{Eu}^{2+}$  ( $4f^7$ ) sublattice. Substitution of half the europium content with  $\text{Gd}^{3+}$  to form  $\text{EuGdCuS}_3$  raises the saturation magnetization to approximately 78 emu/g, consistent with the additive  $\text{Gd}^{3+}$  spin-only contribution expected from site-selective

incorporation. It produces a broader, more field-tunable magnetization response. These results demonstrate that lanthanide sublattice composition provides direct compositional control over the magnetic response of these nanocrystalline materials and constitute the first colloidal synthesis and magnetic characterization of phase-pure  $\text{EuGdCuS}_3$  nanorods. These results constitute the first colloidal synthesis and magnetic characterization of phase-pure  $\text{EuGdCuS}_3$  nanorods and establish a scalable synthetic framework for tunable f-electron-based magnetic nanocrystals.

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## INTRODUCTION

### CHAPTER 1: INTRODUCTION

#### 1.1 Why Nano? Disruptive Technologies and the Nanoscale

From mobile devices to semiconductors, superconductors, and energy conversion systems, modern technologies increasingly demand smaller functional materials, where conventional bulk designs impose direct penalties on cost, efficiency, and practicality. Nanoparticles are the enabling factor within this technological transition. A nanoparticle is defined as a nano-object with all external dimensions in the nanoscale, where the nanoscale spans approximately 1–100 nm.<sup>[1]</sup> Within this size regime, properties bridge molecular and bulk electronic descriptions that can differ significantly.

Nanoparticles are of scientific interest because their controllable size regimes allow for tunable optical and electronic properties; the analogous potential for size-controlled magnetic tunability remains comparatively underexplored and motivates the present work. At least three relevant size regimes merit distinction: bulk, nanoscale, and quantum-confined, the last of which is particularly consequential for semiconductors. In the bulk regime, particle dimensions exceed roughly  $>100$  nm, and physical properties closely resemble those of the macroscopic material; surface contributions and quantum confinement are negligible. As dimensions decrease into the nanoscale regime, particle size becomes comparable to characteristic physical length scales such as the exciton Bohr radius, magnetic correlation length, or electron mean free path, and properties can diverge from bulk behavior. In the quantum-confined regime, when at least one dimension is reduced to a few nanometers, electronic energy levels become discrete rather than continuous, producing size-dependent band gaps and modified magnetic properties.<sup>[2]</sup>

The anisotropic regime, which is the focus of this thesis, arises when different particle dimensions fall into different size regimes simultaneously. In nanorods, for example, the rod length may remain in the bulk-like regime while the diameter falls within the quantum-confined regime, yielding direction-dependent optical, electronic, and magnetic properties that are absent in isotropic particles.

Magnetic materials represent a particularly consequential target for nanoscale engineering. Bulk ferromagnets are composed of magnetic domains whose cooperative alignment produces macroscopic magnetization; reducing a particle below the single-domain threshold eliminates this domain structure and can result in superparamagnetism or ferrimagnetic behavior. The magnetic anisotropy energy, which governs spin-alignment barriers, is strongly size-dependent at the nanoscale and can be further tuned through compositional design. These size-structure-property relationships motivate the synthesis of magnetic nanocrystals with precise control over both morphology and chemical composition.

Although the colloidal-nanocrystal literature on lanthanide–copper chalcogenides is nascent, there are two papers focusing on synthesis and on optical and photocatalytic applications, such as the use of lanthanide sulfide nanocrystals as photocatalysts for hydrogen evolution.<sup>[10][11]</sup> The magnetic properties of this family at the nanoscale remain essentially unexplored, despite the well-established richness of their bulk magnetic behavior. This thesis, therefore, takes a deliberately different vantage point: rather than treating the Ln–Cu–S framework as an optical platform, it treats the two cation sublattices as independent compositional handles on the magnetic ground state. The central experimental claim is that swapping the occupant of a single crystallographic site, from formally nonmagnetic  $\text{Eu}^{3+}$  to a second spin-active  $S = 7/2$  lanthanide, is sufficient to reshape the bulk magnetic response of the

nanocrystal, and that this level of sublattice-selective compositional control is accessible through a well-designed colloidal synthesis.

## **1.2 Synthesis of Metal Chalcogenide Nanomaterials: Size and Phase Control**

Nanoparticle formation is governed by the balance between thermodynamic and kinetic factors during synthesis. Under thermodynamic control, particles evolve toward structures that minimize the system's total free energy, typically producing equilibrium crystal structures and surface configurations with the lowest surface energy. Under kinetic control, which governs most colloidal syntheses, reaction pathways and growth rates determine the final particle morphology before the thermodynamic minimum is reached; metastable structures can form because the system becomes trapped in local energy minima. Reaction parameters such as temperature, precursor concentration, and ligand interactions can shift the system between these regimes, influencing both shape and crystal structure.<sup>[4]</sup>

The nucleation and growth processes that govern nanoparticle formation are commonly described using the LaMer model, which proposes that synthesis proceeds through a rapid burst of nucleation once monomer supersaturation exceeds a critical threshold, followed by diffusion-controlled growth as the monomer concentration decreases. Temporal separation of nucleation and growth limits the number of particles formed during growth, thereby promoting narrower size distributions.<sup>[3]</sup> Precise control over nanoparticle size is consequently essential for tailoring material performance, since dimensions approaching characteristic physical length scales determine whether a given property is bulk-like or quantum-modified.

Phase identity adds a second axis of complexity beyond size. In copper sulfide systems, for example, the  $\text{Cu}_{2-x}\text{S}$  phase diagram contains many closely related polymorphs, chalcocite,



djurleite, roxbyite, digenite, and covellite, whose compositions differ by less than five atomic percent copper. Adjacent phases display distinct electronic and optical signatures, and their interconversion can be monitored through stoichiometry-dependent plasmonic responses. Because phase identity depends sensitively on the copper-to-sulfur ratio, even minor deviations in precursor stoichiometry can shift the product across a phase boundary.<sup>[5]</sup> These considerations are amplified in ternary and multinary chalcogenide systems, where an additional cation sublattice introduces a second compositional variable whose influence on phase stability can be stronger than, and less well documented than, its binary analog.

## **1.2 Anisotropic Growth**

Anisotropic nanostructures, such as nanorods, arise from symmetry-breaking during crystal growth. This symmetry-breaking can be driven by structural features intrinsic to the crystal lattice, such as the existence of crystallographically inequivalent facets with different surface energies, or by external kinetic effects including twin defects, preferential ligand adsorption on specific faces, or asymmetric monomer deposition on seed particles. In both cases, growth is preferentially along one or more crystallographic directions, yielding elongated morphologies with direction-dependent properties.<sup>[4]</sup>

Shape matters beyond mere aesthetics. Different crystal facets expose surface atoms with distinct coordination environments, which influence catalytic reactivity and adsorption behavior. The optical phenomenon of localized surface plasmon resonance depends strongly on particle geometry, because the polarization and spatial distribution of surface electrons vary with shape. Nanorods exhibit distinct plasmon modes along their longitudinal and transverse axes that are absent in spherical nanoparticles. In magnetically ordered systems, the magnetic shape

anisotropy of elongated particles can modify coercive fields and domain behavior, making morphological control a lever for engineering macroscopic magnetic response. Because anisotropic nanostructures thus access property spaces unavailable to isotropic particles, controlling anisotropic growth during synthesis has become an important objective in functional nanocrystal design.

In the  $\text{Eu}_2\text{CuS}_3$  system, the orthorhombic layered structure provides an intrinsic structural basis for anisotropic nanorod growth. The  $\text{CuEu}(1)\text{S}_3$  layers perpendicular to the c-axis create anisotropic surface energies that favor preferential elongation along a specific crystallographic direction, a tendency further modulated by the iodide ions introduced through the  $\text{CuI}$  precursor, which selectively adsorb on particular crystal faces to direct rod-like morphology.<sup>[11]</sup>

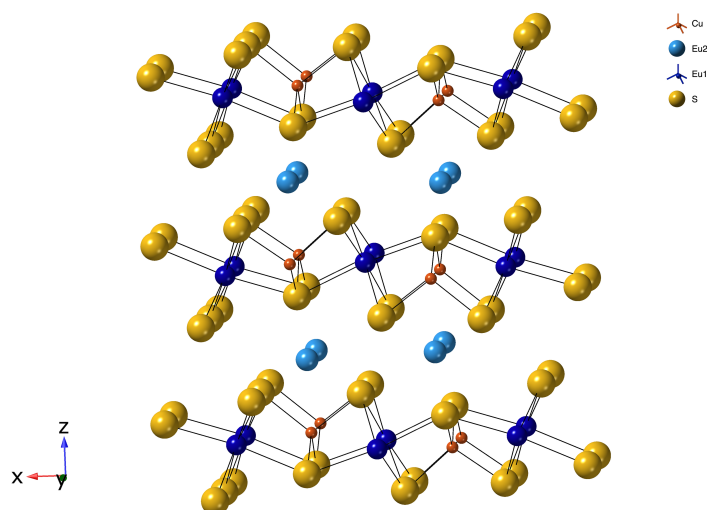
### **1.3 Ternary Chalcogenides as a Composition Handle for Magnetism**

Ternary chalcogenides introduce an additional cation sublattice, expanding the design space beyond what binary systems can access. The presence of two distinct metal species affords greater flexibility in coordination environments, cation ordering, and lattice symmetry, enabling more direct control over electronic and magnetic properties through compositional tuning.<sup>[6]</sup> In contrast to binary systems, where narrow stoichiometric stability windows constrain properties, the expanded compositional freedom of ternary compounds allows properties to be engineered by selecting appropriate combinations of metal ions.

The lanthanide series is particularly well-suited to this compositional engineering strategy for magnetic applications. The 4f electrons of trivalent lanthanides sit beneath filled  $5s^2$  and  $5p^6$  shells, which shield them from ligand field perturbations. As a result, crystal field effects are orders of magnitude smaller than spin–orbit coupling and interelectronic repulsion, and the

magnetic ground state is determined primarily by the atomic 4f configuration according to Hund's rules.<sup>[8]</sup> The  $\text{Ln}^{3+}$  oxidation state is stable across most of the lanthanide series because ionization removes 6s/5d electrons rather than 4f electrons, leaving the magnetic configuration intact. The lanthanide contraction further allows size-matching of specific lanthanide ions to available crystallographic sites in host frameworks, with  $\text{Ln}^{3+}$  radii spanning approximately 1.03 Å (La) to 0.86 Å (Lu) in eight-coordination.

$\text{Eu}_2\text{CuS}_3$  crystallizes in the orthorhombic space group Pnma ( $\text{Eu}_2\text{CuS}_3$ -type structure). In this framework, the Eu(1) site (dark blue, Fig. 1) is coordinated by six sulfur ions, and the  $\text{Eu}(1)\text{S}_6$  octahedra share corners along the a-axis and edges along the b-axis. These octahedra and  $\text{CuS}_4$  tetrahedra together form  $\text{CuEu}(1)\text{S}_3$  layers perpendicular to the c-axis. The Eu(2) ions (light blue, Fig. 1) are coordinated by seven sulfur ions between these layers, forming  $\text{EuS}_7$  monocapped trigonal prisms. The formal oxidation state assignment is  $\text{Eu}^{2+}\text{Eu}^{3+}\text{Cu}^+\text{S}_3^{2-}$ , with  $\text{Eu}^{2+}$  at the Eu(2) site and  $\text{Eu}^{3+}$  at the Eu(1) site.<sup>[10]</sup>



**Figure 1.** Crystal structure of  $\text{Eu}_2\text{CuS}_3$  viewed along the b-axis, illustrating the orthorhombic Pnma framework.  $\text{Eu}(1)\text{S}_6$  octahedra (dark blue) share corners along the a-axis and edges along the b-axis, and interconnect with  $\text{CuS}_4$  tetrahedra (red/Cu) to form  $\text{CuEu}(1)\text{S}_3$  layers perpendicular to the c-axis. Eu(2) cations (light blue) occupy monocapped trigonal prismatic  $\text{EuS}_7$  sites

*between the layers. Sulfur atoms are shown in yellow. The mixed-valent assignment places  $\text{Eu}^{3+}$  at the octahedral Eu(1) site and  $\text{Eu}^{2+}$  at the seven-coordinate Eu(2) site.*

The two europium oxidation states present strikingly different magnetic characters. The  $\text{Eu}^{2+}$  ion carries a large spin-only moment ( $4f^7$ ,  $S = 7/2$ , theoretical moment  $7.94 \mu_B$ ), isoelectronic with  $\text{Gd}^{3+}$ , which couples ferromagnetically through superexchange via the sulfide lattice to produce long-range magnetic ordering at low temperatures. The other europium oxidation state,  $\text{Eu}^{3+}$  ( $4f^6$ ), presents a striking contrast: the  $^7F_0$  Hund's-rule ground state is formally nonmagnetic ( $J = 0$ ), but low-lying  $^7F_J$  ( $J = 1, 2, \dots$ ) excited states lie close enough in energy, on the order of  $k_B T$  at room temperature, that thermal population and Zeeman admixture yield a temperature-dependent Van Vleck paramagnetism rather than a strict diamagnetic response. In practical terms, however, this Van Vleck contribution is negligible for long-range magnetic ordering in  $\text{Eu}_2\text{CuS}_3$ : the  $\text{Eu}^{3+}$  moments are too small and, more importantly, too weakly coupled to participate in cooperative exchange with the  $\text{Eu}^{2+}$  sublattice. Because  $\text{Eu}^{2+}$  and  $\text{Eu}^{3+}$  occupy crystallographically distinct sites, only the Eu(2) sublattice is magnetically active for ordering purposes, and bulk  $\text{Eu}_2\text{CuS}_3$  accordingly exhibits a ferromagnetic transition at approximately 3.4 K arising from interactions among the  $\text{Eu}^{2+}$  ions alone.<sup>10]</sup>

When half of the europium is replaced by  $\text{Gd}^{3+}$  to form  $\text{EuGdCuS}_3$ , the compound retains the  $\text{Eu}_2\text{CuS}_3$ -type orthorhombic structure (space group  $\text{Pnma}$ ) for  $\text{R} = \text{Gd–Dy}$ , since the  $\text{Gd}^{3+}$  ionic radius ( $0.938 \text{ \AA}$  in six-coordination) is compatible with the Eu(1) octahedral site. The lattice parameters contract continuously as the  $\text{R}^{3+}$  ionic radius decreases across the lanthanide series, with no discontinuous change until the  $\text{R} = \text{Tm–Lu}$  range, where a transition to the  $\text{KZrCuS}_3$ -type structure (space group  $\text{Cmcm}$ ) occurs.<sup>[10]</sup> For  $\text{EuGdCuS}_3$ , the  $\text{Gd}^{3+}$  ion effectively replaces the previously diamagnetic  $\text{Eu}^{3+}$  at the Eu(1) site, introducing a second magnetically active sublattice with  $S = 7/2$ .

This mixed-valent character of europium motivates the use of  $\text{Gd}^{3+}$  as a dopant. The presence of crystallographically distinct  $\text{Eu}^{2+}$  and  $\text{Eu}^{3+}$  sites within the same framework is particularly attractive for targeted substitution: the two sites differ in coordination number, geometry, and ionic radius, meaning that a dopant ion can be directed to one site selectively based on its own size and valence, leaving the other site and its magnetic character intact. Gadolinium(III) has a  $[\text{Xe}] 4f^7$  configuration yielding an  $^8S_{7/2}$  free-ion ground state with  $L = 0$  and  $J = S = 7/2$ . The vanishing orbital angular momentum makes  $\text{Gd}^{3+}$  a purely spin-only ion with a theoretical moment of  $7.94 \mu_B$ , the highest of any trivalent lanthanide. The isotropic S-state ground level is split negligibly by the crystal field, so the measured moment closely approaches the free-ion value regardless of the host structure.<sup>[9]</sup> This isotropy is analytically important: if  $\text{Gd}^{3+}$  is successfully incorporated into the  $\text{Eu}_2\text{CuS}_3$  lattice at the Eu(1) site previously occupied by the diamagnetic  $\text{Eu}^{3+}$ , the result is a new magnetic sublattice whose contribution to total magnetization is quantifiable and predictable without the orientation-dependent anisotropy barriers that complicate analysis for other high-moment lanthanides such as  $\text{Dy}^{3+}$  or  $\text{Tb}^{3+}$ .

The bulk  $\text{EuRCuS}_3$  series ( $R = \text{Y, Gd-Lu}$ ), first synthesized and characterized by solid-state methods, demonstrates the breadth of magnetic behavior accessible through lanthanide sublattice selection. Members with non-magnetic R ions ( $R = \text{Y, Lu}$ ) show ferromagnetic ordering of the  $\text{Eu}^{2+}$  sublattice near 5 K. Members with magnetic R ions ( $R = \text{Gd, Tb, Dy, Tm}$ ) show ferrimagnetic transitions near 5 K arising from antiferromagnetic coupling between the  $\text{Eu}^{2+}$  and  $\text{R}^{3+}$  sublattices.  $\text{EuGdCuS}_3$  in particular exhibits an L-type ferrimagnetic transition at approximately 5.4 K, reflecting the antiferromagnetic coupling between two  $S = 7/2$  sublattices with slightly different ordering temperatures.<sup>[10]</sup> Crucially, this site-selective compositional

tuning has been demonstrated only by high-temperature solid-state synthesis, which offers thermodynamic equilibration but no control over particle size, morphology, or surface structure. Achieving the same site-selective substitution through colloidal chemistry, where the synthesis is kinetically controlled, growth is confined to the nanoscale, and the lanthanide ions must be co-delivered to the growing crystal at compatible rates, is a distinct and unsolved problem. The nanoscale behavior of the  $\text{EuRCuS}_3$  family, including whether bulk magnetic ordering persists under reduced dimensionality and an increased surface-to-volume ratio, has not been reported. Addressing both questions simultaneously, whether colloidal synthesis can reproduce the sublattice-selective compositional control achieved in the solid state, and whether the resulting nanocrystals retain the characteristic magnetic behavior of the bulk, is the primary gap this work seeks to fill.

#### **1.4 Synthetic Strategy: SSP Approach and HSAB Considerations**

The single-source precursor (SSP) strategy is adopted because it pre-assembles metal–sulfur bonds within a single molecular entity prior to decomposition, thereby avoiding the differential reactivity and local stoichiometry fluctuations that arise when separate metal and chalcogen sources react simultaneously. This consideration is especially important in the ternary  $\text{Ln–Cu–S}$  system, where two metals with very different Lewis acid characters must be delivered to the growing nanocrystal at controlled relative rates. Dithiocarbamate (DTC) ligands are the SSP of choice: they are readily synthesized, air-stable, and form isolable complexes across a wide range of metal oxidation states.<sup>[12]</sup>

The hard–soft acid–base (HSAB) framework underpins both ligand design and copper source selection.  $\text{Cu}^+$  (soft acid) forms strong, thermodynamically stable bonds with the DTC

sulfur donors (soft base), consistent with the phase stability of the Cu–S product.  $\text{Ln}^{3+}$  ions are hard acids and intrinsically less suited to sulfur ligation; however, the chelate effect of the bidentate DTC provides sufficient thermodynamic stabilization to maintain the Ln–DTC complex through the synthesis temperature window. The hard–soft mismatch nonetheless creates a phase segregation risk: if Ln–S binding is insufficiently robust relative to competing Ln–O pathways,  $\text{Ln}^{3+}$  ions may segregate to the nanocrystal surface or form oxysulfide impurities, particularly in the presence of trace oxygen. This risk is managed through a strict inert-atmosphere technique and careful precursor drying and degassing. The copper source selection also follows HSAB reasoning: CuI provides  $\text{Cu}^+$  (soft acid), which is well matched to sulfur donors, whereas  $\text{Cu}(\text{acac})_2$  provides  $\text{Cu}^{2+}$  (harder acid), whose reduction during synthesis introduces kinetic complications that can divert the system toward competing phases.

The compatible decomposition kinetics of  $\text{Eu}(\text{DTC})_3(\text{phen})$  and  $\text{Gd}(\text{DTC})_3(\text{phen})$  are expected to enable concurrent delivery of both lanthanides to the growing nanocrystal. The heterogeneous nucleation mechanism operative in these systems, wherein  $\text{Cu}_{2-x}\text{S}$  nanoparticles form first at lower temperatures and then serve as seeds into which Ln monomers diffuse and incorporate upon further heating, has been established in the closely related  $\text{CuLnS}_2$  and  $\text{CuEu}_2\text{S}_3$  systems.<sup>[11]</sup> This two-stage nucleation pathway bypasses the high activation-energy barriers to homogeneous  $\text{Ln}_2\text{S}_3$  nucleation, which would otherwise require temperatures exceeding the practical limits of wet-chemistry synthesis.

### **1.5 Copper-to-Sulfur and Lanthanide-to-Copper Ratio Control**

In binary copper sulfide synthesis, the Cu:S ratio is the primary determinant of phase identity. Adjacent phases in the  $\text{Cu}_{2-x}\text{S}$  system are separated by copper-composition windows of

less than five atomic percent, meaning that small deviations in precursor stoichiometry can shift the product across a phase boundary entirely.<sup>[5]</sup> Beyond phase identity, the Cu:S ratio determines the copper vacancy concentration, which governs free-hole density and thereby introduces a stoichiometry-dependent Pauli paramagnetic background in magnetization measurements. Reproducible control over the binary Cu:S ratio is therefore a prerequisite for unambiguous interpretation of magnetic data in doped systems.

In ternary synthesis, the Ln:Cu ratio introduces a second compositional variable. Even modest lanthanide loadings can alter nucleation thermodynamics, destabilize the target Cu–S phase, or create solubility limits beyond which  $\text{Ln}^{3+}$  segregates as a binary sulfide impurity rather than incorporating into the host lattice. The sensitivity of ternary Cu-based chalcogenide phase identity to secondary cation content has been demonstrated in the analogous Na–Cu–S system, where precursor choice and stoichiometry together determine whether thermodynamic or metastable polymorphs form.<sup>[7]</sup> In the  $\text{EuGdCuS}_3$  system specifically, the relevant crystal-chemistry question is whether  $\text{Gd}^{3+}$  can be incorporated at the six-coordinate Eu(1) site that  $\text{Eu}^{3+}$  formally occupies in  $\text{Eu}_2\text{CuS}_3$ . The Shannon radii of the two trivalent ions in six-coordination are nearly identical, 0.938 Å for  $\text{Gd}^{3+}$  and 0.947 Å for  $\text{Eu}^{3+}$ , so size compatibility at the target site is excellent, and lattice strain upon substitution is expected to be minimal. The other europium site in this structure, the seven-coordinate Eu(2) occupied by  $\text{Eu}^{2+}$ , has a much larger radius (~1.20 Å for  $\text{Eu}^{2+}$  in 7-coordination), and substitution there would be prohibitively strained. The relevant ionic-radius comparison for doping into  $\text{Eu}_2\text{CuS}_3$  is therefore  $\text{Gd}^{3+} \rightarrow \text{Eu}^{3+}$  at Eu(1), not  $\text{Gd}^{3+} \rightarrow \text{Eu}^{2+}$  at Eu(2). The substitution preserves charge balance, since both  $\text{Gd}^{3+}$  and  $\text{Eu}^{3+}$  carry a +3 charge. Despite this favorable crystal-chemistry picture, PXRD must independently confirm phase purity for each composition, since kinetic factors during colloidal synthesis, competing



nucleation of binary Gd–S and Eu–S phases, differential precursor decomposition rates, and segregation to the nanocrystal surface, can still divert the system away from the target ternary even when the substitution is thermodynamically accessible.

## CHAPTER 2: SYNTHESIS AND CHARACTERIZATION METHODS

### 2.1 Materials and Instrumentation

La(III) nitrate hexahydrate (99.999%, Sigma), Ce(III) nitrate hexahydrate (99.99%, Sigma), Pr(III) nitrate hexahydrate (99.9%, Sigma), Nd(III) nitrate hexahydrate (99.9%, Sigma), Sm(III) nitrate hexahydrate (99.9%, Sigma), Eu(III) nitrate hexahydrate (99.9%, ThermoFisher), Gd(III) nitrate hexahydrate (99.9%, Sigma), Tb(III) nitrate hexahydrate (99.9%, Sigma), Dy(III) nitrate hexahydrate (99.9%, Sigma), Ho(III) nitrate hexahydrate (99.9%, Sigma), Er(III) nitrate hexahydrate (99.9999%, Sigma), Tm(III) nitrate hexahydrate (99.9%, Sigma), Yb(III) nitrate hexahydrate (99.9%, Sigma), Lu(III) nitrate hexahydrate (99.9%, Alfa Aesar), Y(III) nitrate hexahydrate (99.99%, REacton), 1,10-phenanthroline (99%, Alfa Aesar), copper(I) iodide (98%, Sigma), copper(II) acetylacetonate (97%, Sigma), Nd(III) chloride (99.99%, anhydrous, Sigma), Nd(III) iodide (99.99%, anhydrous, Sigma), Gd(III) chloride (99%, anhydrous, Sigma), Gd(III) iodide (99.99%, anhydrous, Sigma), carbon disulfide (99%, Sigma), lithium bis(trimethylsilyl)amide (Sigma), ammonium diethyldithiocarbamate (97%, Sigma), acetonitrile (99.8%, Sigma), ethanol (200 proof, Warner Graham), hexanes ( $\geq 99\%$ , mixture of isomers, Sigma) were used as received. Oleylamine (70%, Sigma) was distilled under vacuum to a clear and colorless liquid prior to use and stored in the glovebox under nitrogen.

Characterization techniques employed in this work include powder X-ray diffraction (PXRD), transmission electron microscopy (TEM) and high-resolution TEM (HRTEM), scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM-EDS), inductively coupled plasma mass spectrometry (ICP-MS), and superconducting quantum interference device (SQUID) magnetometry. Phase assignments were made by comparison with reference patterns from the PDF-4+ database.

## 2.2 Synthesis of Phase-Pure Lanthanide Sulfide Nanoparticles: $\text{Nd}_3\text{S}_4$ and $\text{Gd}_3\text{S}_4$

Binary lanthanide sulfide syntheses were conducted to establish phase-accessible conditions across varying precursor identities ( $\text{LnCl}_3$  vs.  $\text{LnI}_3$ ) and sulfur delivery strategies ( $\text{CS}_2$  alone vs.  $\text{CS}_2$  with LiHMDS activation), to identify routes to phase-pure  $\text{Ln}_3\text{S}_4$  ( $\text{Ln} = \text{Nd}, \text{Gd}$ ) nanoparticles. Under glovebox conditions ( $\text{N}_2$ ),  $\text{LnX}_3$  ( $\text{Ln} = \text{Nd}, \text{Gd}$ ;  $\text{X} = \text{Cl}, \text{I}$ ; 0.13 mmol) was added to oleylamine (6 mL) in a round-bottom flask. Carbon disulfide (0.4 mL) was then introduced, producing a gel-like mixture with a noticeable exotherm. The mixture was degassed at room temperature (25 °C) under vacuum for approximately 20 minutes. A sand bath preheated to 300 °C was prepared, and the degassed flask was transferred into it and heated under a continuous flow of argon for 1 hour. After cooling for approximately 10 minutes under ambient conditions, the crude product was washed twice with hexanes (5 mL) and ethanol (40 mL), sonicating and centrifuging at 4500 rpm for 5 minutes at each step. The final product was redispersed in approximately 5 mL hexanes and stored under inert atmosphere.

In an alternative synthesis route, lithium bis(trimethylsilyl)amide (LiHMDS; 0.65 mmol) was added with all other reagents prior to degassing and heating. LiHMDS serves to deprotonate oleylamine, generating a more reactive amide nucleophile that activates  $\text{CS}_2$  toward nucleophilic insertion and controlled C–S bond formation, thereby facilitating sulfur delivery and promoting Ln–S bond formation under milder solution-phase conditions. This approach was particularly relevant for  $\text{NdI}_3$ , where the higher Nd–I lattice energy benefits from enhanced sulfur precursor reactivity.

### 2.3 Synthesis of $\text{Ln}(\text{DTC})_3(\text{phen})$ Single-Source Precursors ( $\text{Ln} = \text{La-Lu}$ )

Ammonium diethyldithiocarbamate ( $\text{Et}_2\text{NCSNH}_4$ , DTC; 6.0 mmol) and 1,10-phenanthroline (phen; 2.0 mmol) were dissolved in ethanol (20 mL) and acetonitrile (20 mL) under stirring. Separately,  $\text{Ln}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$  (2.0 mmol) was dissolved in ethanol (10 mL). The metal salt solution was added dropwise to the ligand solution under continuous stirring, whereupon the mixture became cloudy with a noticeable color change. After stirring at room temperature for 1 hour, the product was isolated by vacuum filtration, washed thoroughly with ethanol, and dried under reduced pressure to yield the SSP as a dry powder, which was stored in a vial under ambient conditions. This precipitation route proved robust and reproducible across all lanthanides from La through Lu.

### 2.4 Synthesis of $\text{Eu}_2\text{CuS}_3$

$\text{Eu}(\text{DTC})_3(\text{phen})$  (0.20 mmol), CuI (0.10 mmol), and oleylamine (8 mL) were combined in a round-bottom flask on a Schlenk line. The mixture was placed under vacuum and heated to 115 °C, then degassed at this temperature for 30 minutes to remove residual moisture and oxygen from the oleylamine and precursor coordination sphere. The atmosphere was switched to argon, and the reaction temperature was raised to 330 °C and maintained for 30 minutes. After cooling under ambient conditions for approximately 10 minutes, the crude product was washed twice with hexanes (5 mL) and ethanol (40 mL) with sonication and centrifugation at 4500 rpm for 5 minutes per wash. The final product was redispersed in approximately 5 mL hexanes and stored under inert atmosphere.

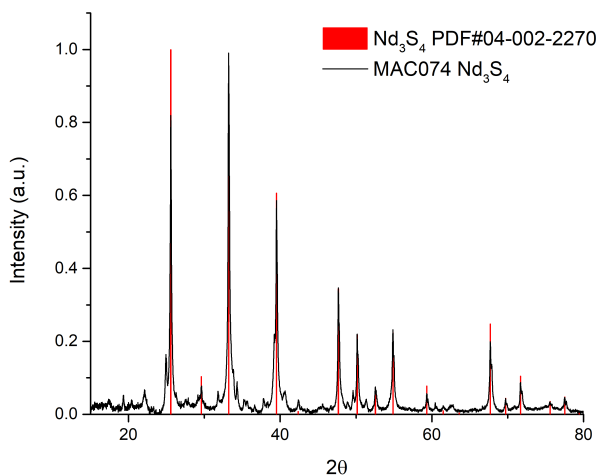
## 2.5 Synthesis of EuGdCuS<sub>3</sub>

Eu(DTC)<sub>3</sub>(phen) (0.10 mmol), Gd(DTC)<sub>3</sub>(phen) (0.10 mmol), CuI (0.10 mmol), and oleylamine (8.0 mL) were combined in a round-bottom flask on a Schlenk line. The mixture was placed under vacuum, heated to 115 °C, and degassed for 45 minutes. The atmosphere was switched to argon, the temperature was raised to 315 °C, and the reaction was maintained at this temperature for 30 minutes. After cooling, the crude product was washed and isolated by the same hexanes/ethanol procedure described above. The final product was redispersed in hexanes and stored under inert atmosphere.

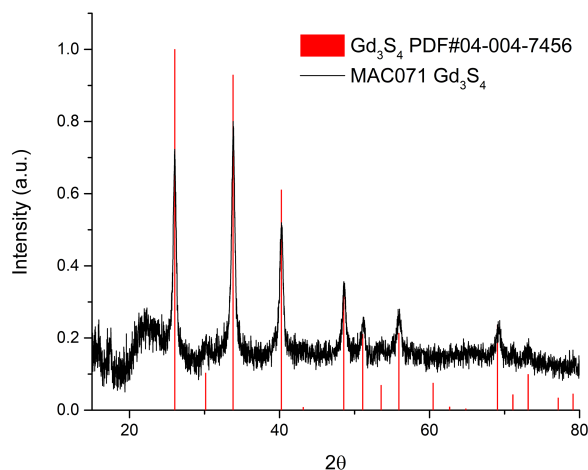
## CHAPTER 3: RESULTS AND DISCUSSION

### 3.1 Phase-Pure $\text{Nd}_3\text{S}_4$ and $\text{Gd}_3\text{S}_4$ Nanoparticles via Precursor Screening

The binary lanthanide sulfide syntheses were conducted via a non-SSP route, reacting  $\text{LnX}_3$  ( $\text{X} = \text{Cl}, \text{I}$ ) directly with  $\text{CS}_2$  in oleylamine, independently of the SSP strategy employed for the ternary syntheses. The reaction of  $\text{LnX}_3$  ( $\text{X} = \text{Cl}, \text{I}$ ) with  $\text{CS}_2$  in oleylamine at 300 °C under argon produced isolable lanthanide sulfide nanoparticles. Phase identity and purity were confirmed by PXRD, with diffractograms for both systems indexing to the  $\text{Th}_3\text{P}_4$ -type  $\text{Ln}_3\text{S}_4$  structure (cubic space group  $I\bar{4}3d$ ), consistent with the reference patterns for  $\text{Nd}_3\text{S}_4$  and  $\text{Gd}_3\text{S}_4$ , respectively. The addition of LiHMDS facilitated sulfur delivery and promoted  $\text{Ln-S}$  bond formation under milder conditions, proving particularly beneficial for  $\text{NdI}_3$ , where the higher lattice energy of the precursor would otherwise require more forcing conditions.



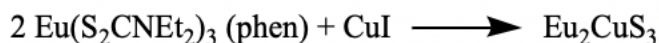
**Figure 2.** PXRD pattern (MAC074, black) of  $\text{Nd}_3\text{S}_4$  nanoparticles synthesized from  $\text{NdX}_3$  ( $\text{X} = \text{Cl}, \text{I}$ ) in oleylamine with  $\text{CS}_2$  at 300 °C under argon, overlaid with the  $\text{Nd}_3\text{S}_4$  reference pattern (PDF#04-002-2270, red). Strong agreement between experimental and reference peak positions confirms phase-pure formation of the  $\text{Th}_3\text{P}_4$ -type cubic structure (space group  $I\bar{4}3d$ ).



**Figure 3.** PXRD pattern (MAC071, black) of  $Gd_3S_4$  nanoparticles synthesized from  $GdX_3$  ( $X = Cl, I$ ) in oleylamine with  $CS_2$  at 300 °C under argon, overlaid with the  $Gd_3S_4$  reference pattern (PDF#04-004-7456, red). Coincidence of experimental and reference peak positions confirms phase-pure formation of the  $Th_3P_4$ -type cubic structure (space group  $I43d$ ).

These binary syntheses confirmed that the SSP-free route using  $CS_2$  as a sulfur source in oleylamine is viable for lanthanide sulfide formation, yielding the thermodynamically accessible  $Th_3P_4$ -type  $Ln_3S_4$  phase under these conditions, and thereby establishing the precursor-chemistry framework for subsequent ternary syntheses. In particular, the successful isolation of phase-pure  $Gd_3S_4$  served as a direct synthetic benchmark for establishing conditions under which  $Gd^{3+}$  can be incorporated into a sulfide lattice via the oleylamine/ $CS_2$  route, confirming that reactive sulfide species generated at 300 °C are sufficient to react with gadolinium precursors to produce the target  $Ln_3S_4$  phase. These conditions provide an interesting comparison to the synthesis of  $EuGdCuS_3$ .

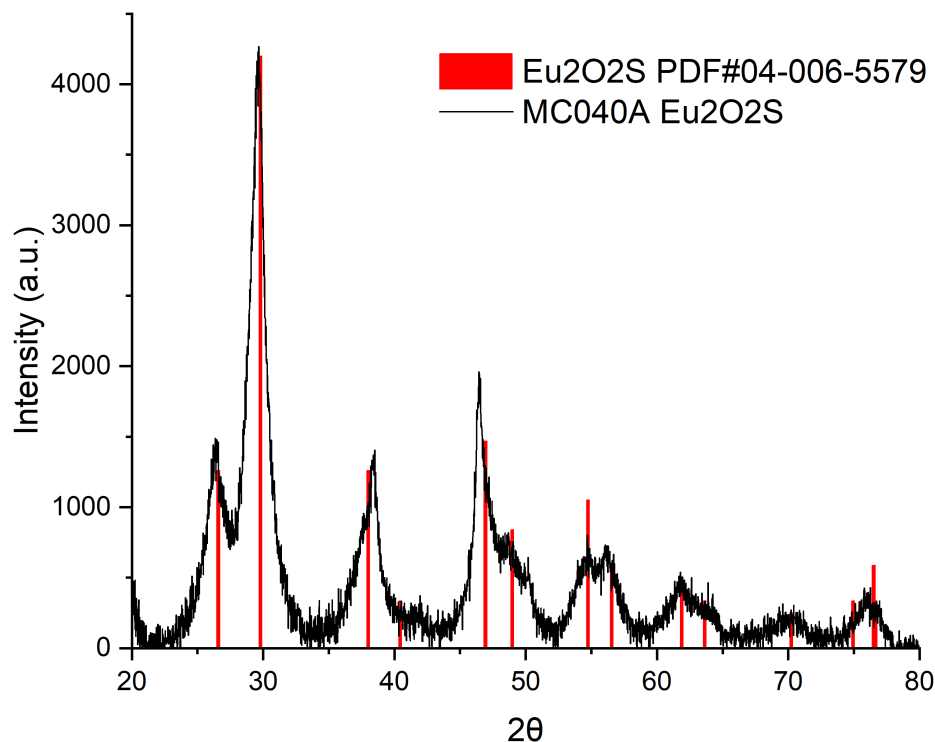
### 3.2 $Eu_2CuS_3$ : Phase-Pure Ternary Nanorod Synthesis and Characterization



Following optimization of the reaction atmosphere and degassing protocol, the reaction of  $\text{Eu}(\text{DTC})_3(\text{phen})$  (0.20 mmol) with  $\text{CuI}$  (0.10 mmol) in oleylamine at 330 °C for 30 minutes

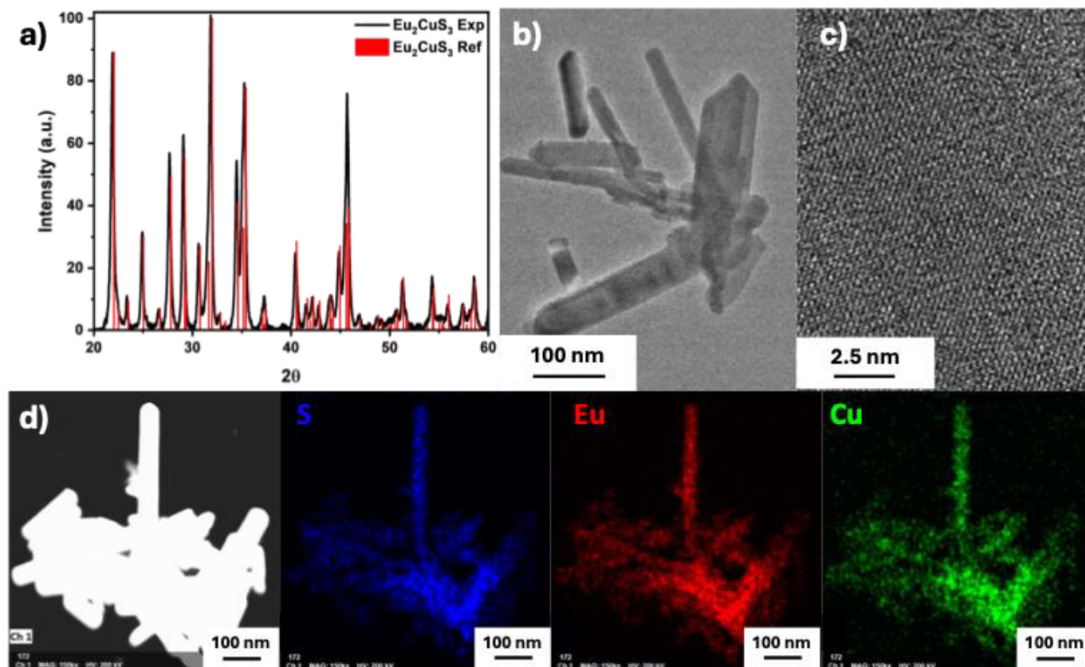
yielded a product whose PXRD pattern (MC042B) matches the  $\text{Eu}_2\text{CuS}_3$  reference (PDF#04-016-1598) with good fidelity, confirming phase-pure ternary product formation. A persistent challenge during early synthesis attempts was the formation of  $\text{Eu}_2\text{O}_2\text{S}$  as an impurity phase, as evidenced by PXRD patterns (MC040A, MC038A, MC038B, MC046) matching PDF#04-006-5579. The appearance of this oxysulfide phase is consistent with air sensitivity and the HSAB mismatch:  $\text{Eu}^{3+}$ , as a hard Lewis acid, has a thermodynamic affinity for hard oxygen donors, and even trace moisture or dissolved oxygen in oleylamine at elevated temperatures is sufficient to drive oxysulfide formation. Mitigation required systematic optimization of degassing temperature, degassing time, and reaction atmosphere, together with the use of freshly distilled oleylamine. Extended degassing at 115 °C under vacuum before heating effectively removed residual water and oxygen from both the solvent and the precursor coordination sphere.





**Figure 4.** PXRD pattern (MC040A) of the crude product obtained from early  $\text{Eu}_2\text{CuS}_3$  synthesis attempts prior to optimization of the degassing protocol. The experimental trace matches the  $\text{Eu}_2\text{O}_2\text{S}$  reference pattern (PDF#04-006-5579), indicating oxysulfide impurity formation consistent with the hard-acid character of  $\text{Eu}^{3+}$  and its thermodynamic affinity for residual oxygen donors in oleylamine at elevated temperature.

The success of  $\text{CuI}$  as the copper source and the difficulties encountered with  $\text{Cu}(\text{acac})_2$  are rationalized through HSAB considerations.  $\text{Cu}^+$  from  $\text{CuI}$  is a soft acid well-matched to the sulfur donors of both the DTC ligand and the growing sulfide lattice, whereas  $\text{Cu}^{2+}$  from  $\text{Cu}(\text{acac})_2$  introduces a harder copper center whose in-situ reduction to  $\text{Cu}^+$  during synthesis adds kinetic complexity that can divert the system toward competing phases or incomplete metal incorporation. Furthermore, consistent with the heterogeneous nucleation mechanism operative in related  $\text{Cu-Ln-S}$  systems, the iodide ion released from  $\text{CuI}$  during the reaction plays an active role in directing anisotropic morphology by selective adsorption on specific crystal faces.<sup>[11]</sup>



**Figure 5.** Characterization of phase-pure  $\text{Eu}_2\text{CuS}_3$  nanorods synthesized from  $\text{Eu}(\text{DTC})_3(\text{phen})$  (0.20 mmol) and  $\text{CuI}$  (0.10 mmol) in oleylamine at 330 °C (30 min). **(a)** PXRD pattern (MC042B, black) overlaid with the  $\text{Eu}_2\text{CuS}_3$  reference pattern (PDF#04-016-1598, red), confirming phase purity. **(b)** Representative low-magnification TEM image showing anisotropic nanorod morphology (scale bar: 100 nm). **(c)** HRTEM image displaying resolved lattice fringes indicative of high crystallinity (scale bar: 2.5 nm). **(d)** HAADF-STEM image and corresponding EDS elemental maps for S (blue), Eu (red), and Cu (green), demonstrating homogeneous co-localization of all three elements across individual nanorods (scale bars: 100 nm).

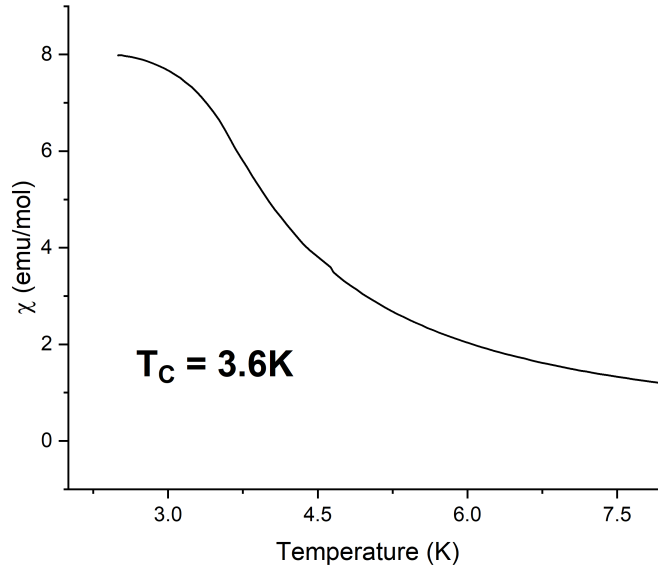
TEM imaging confirms an anisotropic nanorod morphology with high crystallinity, as evidenced by HRTEM lattice fringes. The rod geometry is consistent with preferential growth along a specific crystallographic axis of the orthorhombic  $\text{Eu}_2\text{CuS}_3$  structure, driven by the anisotropic surface energies of the layered framework amplified by iodide-directed growth. EDS elemental mapping demonstrates co-localization of S, Eu, and Cu signals across individual rods, confirming homogeneous elemental distribution within the nanostructures rather than phase-segregated domains.

### 3.2.1 Magnetic Characterization of $\text{Eu}_2\text{CuS}_3$ Nanorods

SQUID magnetometry of  $\text{Eu}_2\text{CuS}_3$  nanorods reveals a ferromagnetic-like hysteresis loop with a saturation magnetization of  $M_{\text{sat}} = 47.8 \text{ emu/g}$  and a Curie temperature of  $T_{\text{C}} = 3.6 \text{ K}$ , in

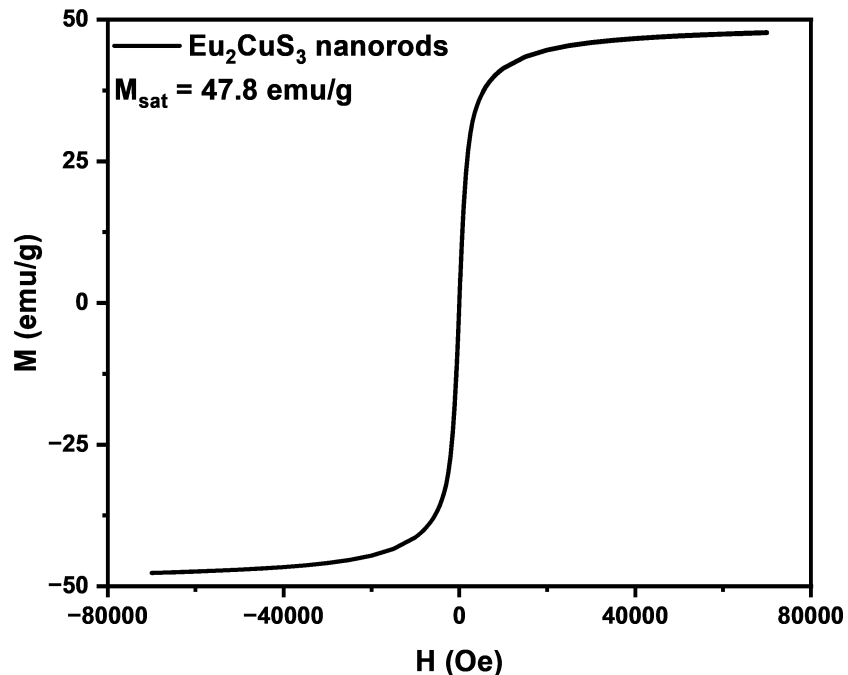
close agreement with the bulk ferromagnetic transition temperature of 3.4 K reported for  $\text{Eu}_2\text{CuS}_3$ .<sup>[10]</sup> The low ordering temperature is consistent with weak f–f exchange interactions mediated by the sulfide lattice, characteristic of  $\text{Eu}^{2+}$ -containing sulfides where the  $4f^7$  configuration carries a large spin moment but couples weakly via superexchange through the Cu 3d and S 3p states. The sharpness of the transition in the  $\chi$  vs. T plot near 3.6 K indicates cooperative magnetic ordering rather than spin-glass freezing, suggesting structural homogeneity and Eu site ordering within the nanorod lattice.

The superexchange pathway responsible for this ordering deserves brief mechanistic comment. Direct  $4f$ – $4f$  overlap between neighboring  $\text{Eu}^{2+}$  ions is negligible because  $4f$  orbitals are spatially contracted and shielded by the filled  $5s^25p^6$  shells. Magnetic coupling therefore proceeds through an indirect pathway in which the  $\text{Eu}^{2+}$   $4f$  spin polarizes bridging S 3p orbitals, which in turn couple through the diamagnetic Cu  $3d^{10}$  sublattice (acting as a ligand bridge rather than a spin carrier) and out through a second S 3p bridge to a neighboring  $\text{Eu}^{2+}$  center. This  $\text{Eu}^{2+} 4f \leftrightarrow \text{S } 3p \leftrightarrow \text{Cu } 3d \leftrightarrow \text{S } 3p \leftrightarrow \text{Eu}^{2+} 4f$  chain is long and involves only filled intermediate orbitals, which accounts for the weakness of the exchange and the correspondingly low ordering temperature. Unambiguous assignment of the specific spin-ordering direction requires neutron diffraction, which is the only technique that directly probes magnetic scattering from ordered sublattices; such measurements have not been performed on  $\text{Eu}_2\text{CuS}_3$  nanorods in this work. The close agreement between the nanorod  $T_C$  (3.6 K) and the bulk value (3.4 K) from Wakeshima's magnetic susceptibility data nonetheless supports the assignment of ferromagnetic ordering of the  $\text{Eu}^{2+}$  sublattice.



**Figure 6.** Temperature-dependent magnetic susceptibility ( $\chi$  vs.  $T$ ) of  $\text{Eu}_2\text{CuS}_3$  nanorods measured by SQUID magnetometry from 2.8 to 7.5 K. The sharp rise in susceptibility indicates a ferromagnetic ordering transition at  $T_C = 3.6$  K, in close agreement with the 3.4 K Curie temperature reported for bulk  $\text{Eu}_2\text{CuS}_3$ . The transition arises from cooperative ordering of the  $\text{Eu}^{2+}$  ( $4f^7$ ,  $S = 7/2$ ) sublattice at the Eu(2) site; the  $\text{Eu}^{3+}$  ( $4f^6$ ,  $J = 0$ ) ions at the Eu(1) site are diamagnetic and do not contribute.

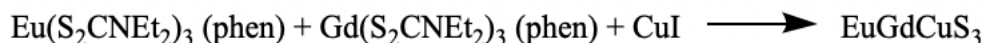
The formal oxidation state of europium in  $\text{Eu}_2\text{CuS}_3$  warrants careful consideration. The nominal stoichiometry is consistent with the mixed-valent assignment  $\text{Eu}^{2+}\text{Eu}^{3+}\text{Cu}^+\text{S}_3^{2-}$ , confirmed by  $^{151}\text{Eu}$  Mössbauer spectroscopy in bulk samples.<sup>[10]</sup> The observed ferromagnetic behavior and substantial  $M_{\text{sat}}$  are most straightforwardly attributed to the magnetically active  $\text{Eu}^{2+}$  ( $4f^7$ ,  $S = 7/2$ ) at the Eu(2) site, since  $\text{Eu}^{3+}$  ( $4f^6$ ,  $J = 0$ ) is diamagnetic. This mixed-valent character motivates the selection of  $\text{Eu}_2\text{CuS}_3$  as the host for  $\text{Gd}^{3+}$  doping at the Eu(1) site, replacing the diamagnetic  $\text{Eu}^{3+}$  with a spin-active dopant.



**Figure 7.** Field-dependent magnetization ( $M$  vs.  $H$ ) isotherm of  $\text{Eu}_2\text{CuS}_3$  nanorods measured by SQUID magnetometry at 2 K over  $\pm 70$  kOe. The sample displays ferromagnetic-like hysteresis with a saturation magnetization of  $M_{\text{sat}} = 47.8$  emu/g, consistent with ordering of the  $\text{Eu}^{2+}$  ( $4f^7$ ) sublattice through sulfide-mediated superexchange.

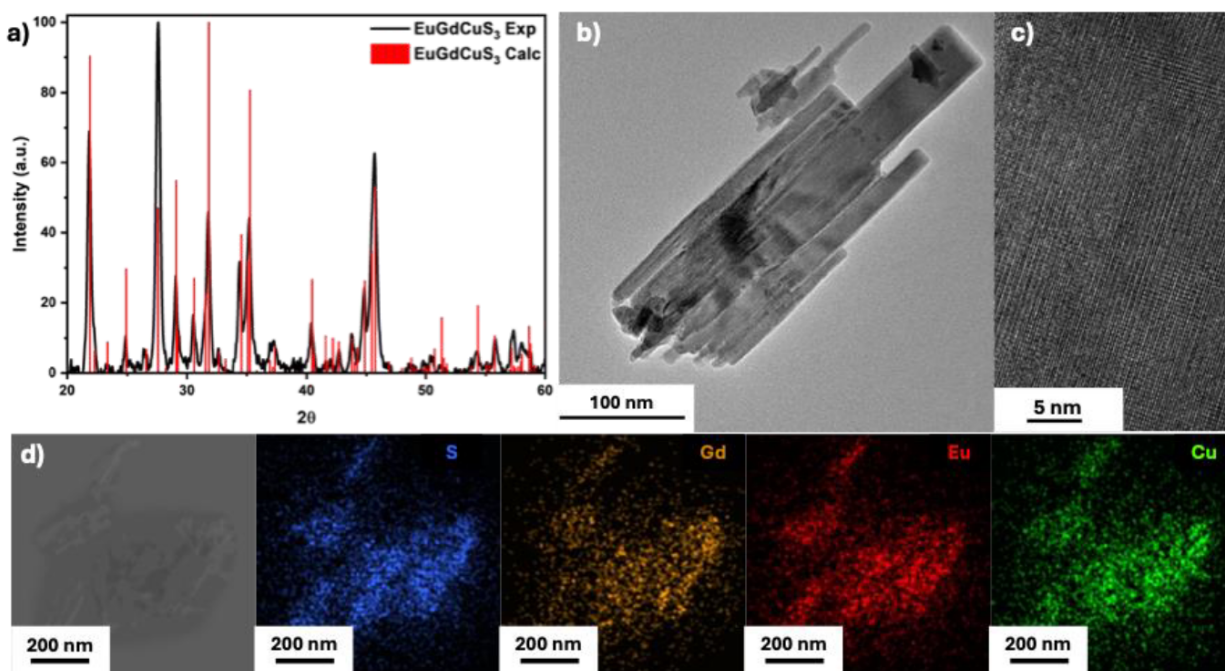
### 3.3 $\text{EuGdCuS}_3$ : Mixed-Lanthanide Ternary Phase, Composition, and Magnetic Tunability

#### 3.3.1 Synthesis and Phase Purity



Substituting half of the europium content with gadolinium by reaction of  $\text{Eu}(\text{DTC})_3(\text{phen})$  (0.10 mmol),  $\text{Gd}(\text{DTC})_3(\text{phen})$  (0.10 mmol), and  $\text{CuI}$  (0.10 mmol) in oleylamine produced a mixed-lanthanide ternary phase whose PXRD pattern (MC048) matches the  $\text{GdEuCuS}_3$  reference (PDF#04-019-4851), confirming that  $\text{Gd}^{3+}$  successfully incorporates into the  $\text{Eu}_2\text{CuS}_3$ -type framework rather than segregating as a binary gadolinium sulfide impurity. The phase-purity result is non-trivial despite the favorable underlying crystal-chemistry.  $\text{Gd}^{3+}$  substitutes into the six-coordinate Eu(1) site, where its Shannon radius (0.938 Å)

is nearly identical to that of  $\text{Eu}^{3+}$  (0.947 Å), so size matching at the target site is excellent, and the isovalent  $\text{Gd}^{3+} \rightarrow \text{Eu}^{3+}$  substitution preserves charge balance. The non-trivial hurdles are kinetic rather than thermodynamic: the two lanthanide precursors must decompose at compatible rates to deliver Eu and Gd to the growing nanocrystal concurrently, and each must incorporate at the correct crystallographic site rather than segregating into separate binary Eu–S or Gd–S nuclei. Phase-pure incorporation, therefore, implies that the SSP decomposition kinetics of  $\text{Eu}(\text{DTC})_3(\text{phen})$  and  $\text{Gd}(\text{DTC})_3(\text{phen})$  are indeed compatible, and that the growing ternary nanocrystal is kinetically competent to absorb both lanthanides at the rates at which they become available. The absence of  $\text{Eu}_2\text{O}_2\text{S}$  impurities in the  $\text{EuGdCuS}_3$  product further confirms that the degassing optimization established for  $\text{Eu}_2\text{CuS}_3$  transfers successfully to the mixed-lanthanide system.



**Figure 8.** Characterization of phase-pure  $\text{EuGdCuS}_3$  nanorods synthesized from  $\text{Eu}(\text{DTC})_3(\text{phen})$  (0.10 mmol),  $\text{Gd}(\text{DTC})_3(\text{phen})$  (0.10 mmol), and  $\text{CuI}$  (0.10 mmol) in oleylamine at 315 °C (30 min). (a) PXRD pattern (MC048, black) overlaid with the calculated  $\text{EuGdCuS}_3$  reference pattern (PDF#04-019-4851, red), confirming phase-pure incorporation of  $\text{Gd}^{3+}$  into the  $\text{Eu}_2\text{CuS}_3$ -type orthorhombic framework ( $Pnma$ ). (b) Low-magnification TEM image showing preserved nanorod morphology (scale bar: 100 nm). (c) HRTEM image revealing resolved lattice fringes and high crystallinity (scale bar: 5 nm). (d) HAADF-STEM image

and corresponding EDS elemental maps for S (blue), Gd (yellow), Eu (red), and Cu (green), demonstrating uniform spatial distribution of all four elements and ruling out phase-segregated Eu-rich and Gd-rich domains (scale bars: 200 nm).

### 3.3.2 Elemental Composition: ICP-MS and EDS

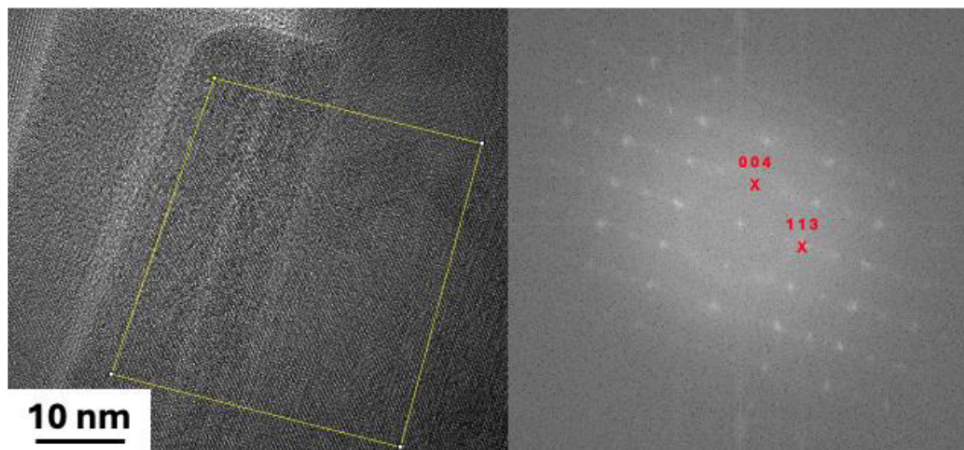
ICP-MS analysis confirmed a Eu:Gd ratio of  $1.00 \pm 0.01$  in the isolated  $\text{EuGdCuS}_3$  product, demonstrating quantitative and stoichiometrically controlled incorporation of both lanthanides. This 1:1 ratio directly mirrors the precursor feed ratio, indicating that neither lanthanide is preferentially excluded from the nanocrystal during growth, a result consistent with the hypothesis that the similar decomposition temperatures of  $\text{Eu}(\text{DTC})_3(\text{phen})$  and  $\text{Gd}(\text{DTC})_3(\text{phen})$  allow both metals to be delivered to the growing particle surface concurrently. EDS elemental mapping shows uniform spatial distribution of S, Gd, Eu, and Cu across the nanoparticle ensemble, consistent with a single homogeneous ternary phase rather than a physical mixture of Eu-rich and Gd-rich domains. The  $\text{Eu}^{2+}+\text{Gd}^{3+}$  mixed-valent composition ( $\text{Eu}^{2+}\text{Gd}^{3+}\text{Cu}^+\text{S}_3^{2-}$ ) is thus confirmed both by diffraction and by compositional analysis.

| Experimental Run                 | Eu-153 Counts (cps) | Gd-158 Counts (cps) |
|----------------------------------|---------------------|---------------------|
| 25 ppb Standard                  | 1325122             | 764167.8            |
| 50 ppb Standard                  | 2509419.8           | 1449919             |
| 75 ppb Standard                  | 3573389.8           | 2075502             |
| 100 ppb Standard                 | 4770854.1           | 2775612             |
| 250 ppb Standard                 | 11760877            | 6843713             |
| EuGdCuS <sub>3</sub> Sample      | 3774404             | 2204184             |
| Analysis of EuGdCuS <sub>3</sub> |                     |                     |
| Concentration of Eu              | 78.2 ± 0.2 ppb      |                     |
| Concentration of Gd              | 78.6 ± 0.7 ppb      |                     |
| Ratio of Eu:Gd = 1.00 ± 0.01     |                     |                     |

**Figure 9.** ICP-MS quantification of the Eu:Gd ratio in isolated  $\text{EuGdCuS}_3$  nanorods. Raw intensities (cps) for  $^{153}\text{Eu}$  and  $^{158}\text{Gd}$  were measured across a calibration series (25, 50, 75, 100, and 250 ppb standards) and for the digested sample. Back-calculated concentrations yield  $[\text{Eu}] = 78.2 \pm 0.2 \text{ ppb}$  and  $[\text{Gd}] = 78.6 \pm 0.7 \text{ ppb}$ , corresponding to a Eu:Gd molar ratio of  $1.00 \pm 0.01$  and confirming stoichiometrically quantitative incorporation of both lanthanides consistent with the 1:1 precursor feed ratio.



HRTEM imaging reveals highly crystalline nanostructures with resolved lattice fringes, and the FFT pattern indexes to the (004) and (113) reflections of the  $\text{EuGdCuS}_3$  structure, confirming long-range crystalline order consistent with the PXRD phase assignment. The nanorod morphology is preserved relative to  $\text{Eu}_2\text{CuS}_3$ , consistent with the structural similarity of the two phases and the dominance of iodide-directed anisotropic growth in both systems.

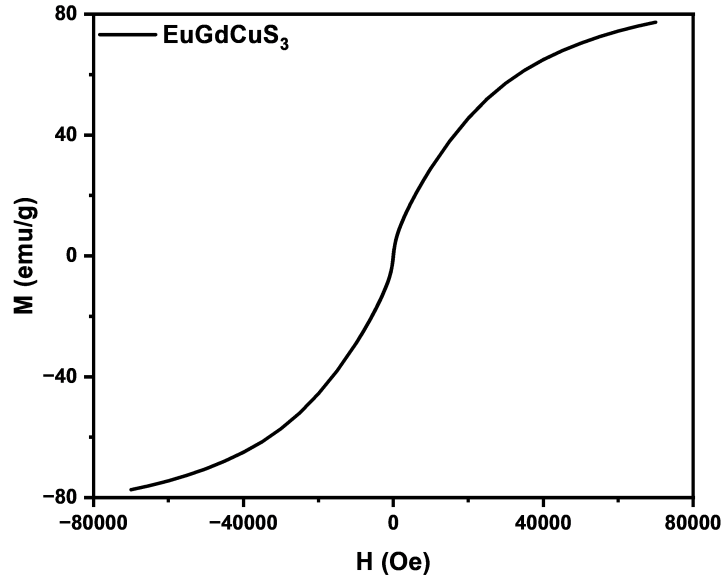


**Figure 10.** HRTEM image of an individual  $\text{EuGdCuS}_3$  nanorod (left; scale bar: 10 nm) and the corresponding fast Fourier transform (FFT) pattern (right) obtained from the boxed region. Reflections in the FFT are indexed to the (004) and (113) planes of the orthorhombic  $\text{EuGdCuS}_3$  structure ( $Pnma$ ), confirming long-range crystalline order consistent with the PXRD phase assignment.

### 3.3.3 Magnetic Properties of $\text{EuGdCuS}_3$

The magnetic consequences of  $\text{Gd}^{3+}$  incorporation are evident in the SQUID data. The  $M$  vs.  $H$  hysteresis loop for  $\text{EuGdCuS}_3$  shows a markedly increased saturation magnetization relative to  $\text{Eu}_2\text{CuS}_3$  (approximately 78 emu/g vs. 47.8 emu/g), consistent with the additive contribution of the  $\text{Gd}^{3+}$  spin-only moment ( $7.94 \mu_B$ ,  $S = 7/2$ ) supplementing the magnetic response of the Eu sublattice. The  $\chi$  vs.  $T$  plot for  $\text{EuGdCuS}_3$  shows a smooth Curie-law-like decay extending to at least 12 K, without the sharp ferromagnetic ordering transition near 3.6 K that characterizes  $\text{Eu}_2\text{CuS}_3$ .



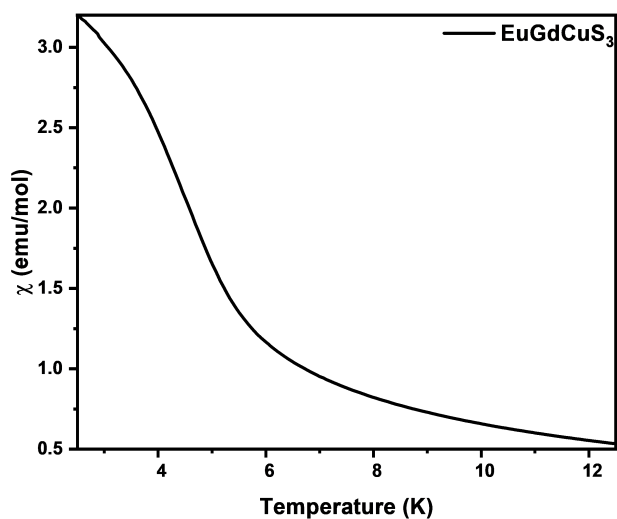


**Figure 11.** Field-dependent magnetization ( $M$  vs.  $H$ ) isotherm of  $\text{EuGdCuS}_3$  nanorods measured by SQUID magnetometry at 2 K over  $\pm 70$  kOe. The saturation magnetization of  $\approx 78$  emu/g represents a  $\sim 63\%$  increase over that of  $\text{Eu}_2\text{CuS}_3$  nanorods (47.8 emu/g), consistent with the additive contribution of the  $\text{Gd}^{3+}$  spin-only moment ( $7.94 \mu_B$ ,  $S = 7/2$ ,  $^8S_{7/2}$ ) supplementing the  $\text{Eu}^{2+}$  sublattice upon substitution of diamagnetic  $\text{Eu}^{3+}$  at the  $\text{Eu}(1)$  site.

This behavioral contrast requires careful interpretation. In bulk  $\text{EuGdCuS}_3$ , the  $\text{Eu}^{2+}$  and  $\text{Gd}^{3+}$  sublattices each carry the  $^8S_{7/2}$  ground state and are antiferromagnetically coupled, producing an L-type ferrimagnetic transition near 5.4 K with a characteristic susceptibility profile including ZFC/FC divergence and a low-temperature drop attributable to partial sublattice compensation.<sup>[10]</sup> In the nanorod system synthesized here, the  $\chi$  vs.  $T$  data show no analogous sharp transition within the measured window of 2.8–12 K. It is tempting to attribute this absence to nanoscale effects, reduced dimensionality, increased surface-to-volume ratio, or broken intersublattice exchange pathways at the nanocrystal surface, and these may well contribute. However, a more candid interpretation is warranted: ferrimagnetic systems do not generally produce a sharp single-feature signature in  $\chi$  vs.  $T$ , and the specific ordering type of a two-sublattice system (ferromagnetic, ferrimagnetic, or antiferromagnetic) cannot be uniquely

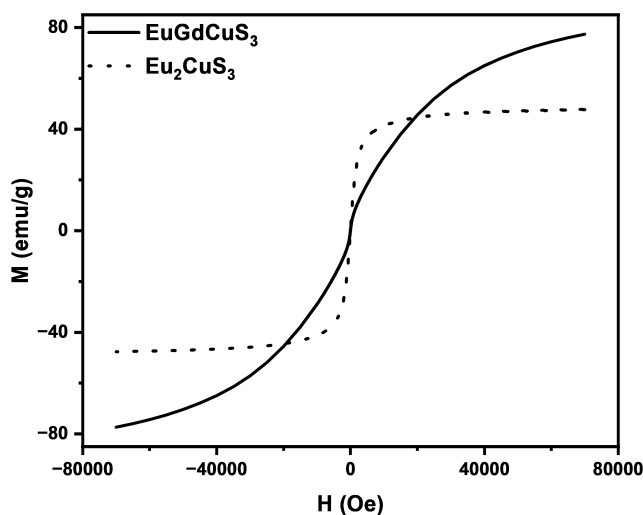
determined from  $\chi$  vs. T and M vs. H alone. Both techniques probe the bulk magnetization response of an ensemble; neither can resolve the relative orientation of the two sublattice moments, which is the defining distinction between ferromagnetic and ferrimagnetic order. Neutron diffraction, which directly probes magnetic scattering from each sublattice, is the technique required to make that assignment unambiguously, and such measurements have not been performed on the nanorods reported here.

What the present data do establish unambiguously is compositional: the 1:1 Eu:Gd stoichiometry confirmed by ICP-MS, the phase purity confirmed by PXRD, the lattice-fringe crystallinity confirmed by HRTEM, and the elemental homogeneity confirmed by EDS together rule out phase-segregated explanations for the magnetic contrast. The observed ~63% increase in saturation magnetization is quantitatively consistent with the additive  $\text{Gd}^{3+}$  spin-only contribution ( $7.94 \mu_B$ ,  $S = 7/2$ ) on top of the  $\text{Eu}^{2+}$  sublattice response, which is the predicted outcome of successful site-selective substitution. The smoother, more sloping approach to saturation in the M vs. H loop of  $\text{EuGdCuS}_3$ , in contrast to the sharp step observed for  $\text{Eu}_2\text{CuS}_3$ , is itself a meaningful signature: it suggests a gradual, field-driven alignment of moments rather than a single cooperative spin-flip event, which is consistent with a tunable magnetic response across a range of applied fields. Extrapolating cautiously from the bulk precedent, the behavior is likely ferrimagnetic in character, but the high saturation magnetization suggests that the  $\text{Eu}^{2+}$  and  $\text{Gd}^{3+}$  sublattices become ferromagnetically aligned at sufficient applied field, canting away from an antiferromagnetic zero-field arrangement. Full resolution of the ordering type ( $T_C$ ) in the nanorod system and the presence or absence of a compensation point require low-temperature magnetometry below 2 K,  $1/\chi$  vs. T Curie–Weiss analysis to extract the Weiss constant, and, ideally, neutron diffraction on the nanorods themselves.



**Figure 12.** Temperature-dependent magnetic susceptibility ( $\chi$  vs.  $T$ ) of  $\text{EuGdCuS}_3$  nanorods measured by SQUID magnetometry from 2.8 to 12 K. The smooth, Curie-law-like decay lacks the sharp ferromagnetic ordering transition observed for  $\text{Eu}_2\text{CuS}_3$  at 3.6 K, suggesting that the ferrimagnetic ordering temperature lies above the measurement window or that reduced dimensionality and enhanced surface-to-volume ratio modify the  $\text{Eu}^{2+}$ – $\text{Gd}^{3+}$  intersublattice exchange coupling relative to bulk  $\text{EuGdCuS}_3$  ( $T_N \approx 5.4$  K).

The overlay comparison of the two  $M$  vs.  $H$  loops makes the magnetization enhancement upon  $\text{Gd}^{3+}$  incorporation visually unambiguous. Taken together, the  $\text{Eu}_2\text{CuS}_3$  and  $\text{EuGdCuS}_3$  results demonstrate that lanthanide sublattice identity directly controls the magnetic ground state of these nanocrystalline materials, in direct analogy with the behavior documented in bulk  $\text{Ln}$ – $\text{Cu}$ – $\text{Q}$  compounds.<sup>[10]</sup>



**Figure 13.** Overlaid  $M$  vs.  $H$  hysteresis loops at 2 K for  $\text{Eu}_2\text{CuS}_3$  (dotted) and  $\text{EuGdCuS}_3$  (solid) nanorods, directly illustrating the magnetization enhancement produced by  $\text{Gd}^{3+}$  substitution at the  $\text{Eu}(1)$  site. The structurally conserved orthorhombic  $\text{Pnma}$  framework and nanorod morphology across both compositions isolate the magnetic contrast to the change in  $f$ -electron sublattice identity, demonstrating that lanthanide selection provides a direct compositional handle on the magnetic ground state of these nanocrystalline materials.

### 3.4 Comparison of $\text{Eu}_2\text{CuS}_3$ vs. $\text{EuGdCuS}_3$ : Structure-Property Relationships

Comparing the two ternary compositions establishes a clear structure–property relationship. Both systems share the  $\text{Eu}_2\text{CuS}_3$ -type orthorhombic framework and nanorod morphology, confirming that the synthetic approach produces structurally analogous phases across the compositional variation. The substitution of  $\text{Gd}^{3+}$  for  $\text{Eu}^{3+}$  at the  $\text{Eu}(1)$  site converts a diamagnetic sublattice into a spin-active one, producing two distinct consequences: a ~63% increase in saturation magnetization (47.8 vs. ~78 emu/g) and a suppression of the sharp ferromagnetic ordering at 3.6 K in favor of a broader, higher-temperature magnetic response. The ICP-MS-confirmed 1:1 Eu:Gd stoichiometry, the phase-pure PXRD pattern, the HRTEM-confirmed crystallinity, and the EDS-confirmed elemental homogeneity collectively rule out phase segregation as an explanation for the magnetic contrast, attributing it instead to the intended change in  $f$ -electron sublattice composition.

One subtlety in the compositional assignment deserves explicit attention. The formal description " $\text{Eu}^{2+}\text{Gd}^{3+}\text{Cu}^+\text{S}_3^{2-}$ " is the site-ordered mixed-valent formulation in which europium is present exclusively as  $\text{Eu}^{2+}$  at the seven-coordinate Eu(2) site and gadolinium occupies the six-coordinate Eu(1) site formerly held by  $\text{Eu}^{3+}$ . An alternative formulation, a  $\text{Eu}^{2+}(\text{Eu}^{3+}_{1-x}\text{Gd}_x)\text{CuS}_3$  solid solution in which  $\text{Gd}^{3+}$  distributes statistically across both lanthanide sites, with europium retaining a mixed  $\text{Eu}^{2+}/\text{Eu}^{3+}$  population, cannot be rigorously excluded by bulk powder diffraction and susceptibility data alone, and Wakeshima's original report does not definitively rule it out. The present work deliberately targeted the site-ordered mixed-valent composition through the choice of a 1:1  $\text{Eu}(\text{DTC})_3(\text{phen})$ :  $\text{Gd}(\text{DTC})_3(\text{phen})$  precursor feed ratio and by structural analogy with  $\text{Eu}_2\text{CuS}_3$ , where the  $\text{Eu}^{3+}$  site is well-defined. All compositional and structural data reported here, the PXRD match to the  $\text{GdEuCuS}_3$  reference pattern (PDF#04-019-4851), the 1:1 Eu:Gd ratio confirmed by ICP-MS, the co-localization of all four elements in EDS maps, and preservation of the  $\text{Eu}_2\text{CuS}_3$ -type framework, are consistent with the mixed-valent site-ordered assignment, but do not by themselves distinguish it from a statistical solid solution. The definitive test is  $^{151}\text{Eu}$  Mössbauer spectroscopy, which resolves  $\text{Eu}^{2+}$  and  $\text{Eu}^{3+}$  environments via their characteristic isomer shifts and would confirm or disprove the mixed-valent formulation directly. Wakeshima and coworkers used this technique to support their bulk assignment, and it is a priority for future work on the nanorod system.<sup>[10]</sup>

These results establish that the SSP colloidal route introduced here provides a general, controllable platform for the engineering of f-electron magnetic nanocrystals. The ability to tune magnetization and ordering behavior through lanthanide sublattice selection within a structurally conserved nanorod morphology opens a design space previously accessible only through high-temperature solid-state synthesis and absent at the nanoscale.

## CHAPTER 4: CONCLUSION

This work reports the first colloidal synthesis of phase-pure  $\text{Eu}_2\text{CuS}_3$  and  $\text{EuGdCuS}_3$  nanorods via a single-source dithiocarbamate precursor strategy in oleylamine. Binary  $\text{Nd}_3\text{S}_4$  and  $\text{Gd}_3\text{S}_4$  nanoparticles were prepared via systematic screening of lanthanide halide precursor identity and sulfur delivery strategy, establishing the conditions for Ln–S bond formation under colloidal conditions and providing a synthetic benchmark for the subsequent ternary work. PXRD confirmed phase purity in all cases, and ICP-MS and EDS established 1:1 Eu:Gd stoichiometry and elemental homogeneity for  $\text{EuGdCuS}_3$ . SQUID magnetometry revealed that  $\text{Eu}_2\text{CuS}_3$  nanorods exhibit a soft ferromagnetic transition at  $T_C = 3.6$  K with  $M_{\text{sat}} = 47.8$  emu/g, in close agreement with the bulk value and consistent with ordering of the  $\text{Eu}^{2+}$  sublattice.  $\text{EuGdCuS}_3$  nanorods show a  $\sim 63\%$  increase in saturation magnetization ( $\sim 78$  emu/g), quantitatively consistent with the additive  $\text{Gd}^{3+}$  spin-only contribution expected from successful site-selective substitution, together with a broader, sloping approach to saturation that suggests a tunable field-dependent magnetic response. The specific spin-ordering type in  $\text{EuGdCuS}_3$  cannot be assigned from  $\chi$  vs.  $T$  and  $M$  vs.  $H$  data alone and remains an open question; definitive resolution awaits low-temperature ( $< 2$  K) magnetometry, Curie–Weiss analysis, and neutron diffraction on the nanorods.

The synthetic strategy developed here, using air-stable, readily prepared  $\text{Ln}(\text{DTC})_3(\text{phen})$  precursors, HSAB-informed selection of the copper source ( $\text{CuI}$ ), and rigorous oxygen-free conditions, provides a scalable framework for the nanoscale synthesis of Ln–Cu–S materials across the full lanthanide series. Future work should address the full  $\text{EuRCuS}_3$  ( $R = \text{Y, Gd–Lu}$ ) phase diagram at the nanoscale; systematic size- and morphology-controlled synthesis as a function of reaction parameters; low-temperature magnetic measurements extending below 2 K

to characterize ferrimagnetic ordering fully;  $^{151}\text{Eu}$  Mössbauer spectroscopy to confirm the mixed-valent  $\text{Eu}^{2+}\text{Gd}^{3+}\text{Cu}^+\text{S}_3^{2-}$  assignment unambiguously; and neutron diffraction on the nanorods to resolve sublattice spin ordering. Extension to heavier chalcogens (Se, Te) would further probe the periodic trends in Ln–chalcogen bonding and electronic structure that motivate the broader research program. ‘

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